

1. A bank wants to predict whether a loan applicant will default based on credit score, income, and past loan history. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect past data with features such as credit score, income and past loan history
 - **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
 - **Feature Selection:** Select features that is dependent and independent variables.
 - **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
 - **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes and other boosting algorithms etc...
 - **Model Training:** Train the data with different models using fit method
 - **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
 - **Make Predictions:** Use the best model to predict loan applicant will default or not.
2. A retail store wants to predict the demand for different products to optimize inventory levels. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect historical data of the products such as weekly or daily sales, product demand
- **Data Preprocessing:** **Data Normalization** and categorical data handling using **label or one-hot encoding**.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict the demands of every product.

3. A factory wants to detect whether a manufactured product is defective based on sensor readings and quality control data. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Get the sensor readings and quality control data
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding**, null values handling.
- **Feature Selection:** Select features that is dependent and independent variables.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes and other boosting algorithms etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to predict the product is defective.

4. A healthcare provider wants to analyze patient symptoms and classify them into different disease categories. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification(Multiple classes)

Step-by-step logic:

- **Data Collection:** Gather patient symptoms and values of heart rate, blood pressure and temperature.
- **Data Preprocessing:** Categorical data handling using **label or one-hot encoding** for symptoms fever(yes or no) column.
- **Feature Selection:** Select features that is dependent and independent variables.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes and other boosting algorithms etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to predict the disease category.

5. An e-commerce company wants to identify and remove fake reviews posted by bots or fraudsters. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect data for real and fake reviews.
- **Text processing:** Tokenize the sentences into separate word, remove stopwords and punctuations
- **Data Preprocessing:** Convert text data into numerical format using TF-IDF vectorizer or word embeddings
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision, recall.
- **Make Predictions:** Use the best model to determine the fake reviews and remove it.

6. A financial firm wants to predict stock price movements based on historical price data and market indicators. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Regression

Step-by-step logic:

- **Data Collection:** Collect historical data of the stock prices and market indicators.
- **Data Preprocessing: Data Normalization** and categorical data handling using **label or one-hot encoding**.
- **Feature Selection:** Selecting feature and target variable(X,Y).
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select regression models such as Multiple linear regression, decision tree, random forest etc...
- **Model Training:** Train the data with different models using **.fit()** method.
- **Model Evaluation:** Evaluate the model using metrics such as **R-squared Score**.
- **Make Predictions:** Use the best model to predict the stock price.

7. A social media platform wants to detect fake user accounts based on user activity and profile data. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect data of account activities such as login type, username, frequency of usage and date of creation.
 - **Data Preprocessing:** Handling missing values and normalize the data.
 - **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
 - **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
 - **Model Training:** Train the data with different models using fit method
 - **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision and recall.
 - **Make Predictions:** Use the best model to predict fake accounts.
8. A marketing agency wants to segment customers into different groups based on their purchasing behavior. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Clustering

Step-by-step logic:

- **Data Collection:** Collect past data with features such as spending score, frequency of purchases.
- **Data Preprocessing:** Data normalization technique and null value handling.
- **Feature Selection:** Selecting feature such as spending score and frequency(X).
- **Model Selection:** Select clustering algorithms such as K-means clustering, affinity propagation, DBSCAN.
- **Determine Optimal Clusters:** Use elbow method and dendrogram to find optimal number of clusters
- **Model Training:** Train the data with different models
- **Analyze Clusters:** interpret and get insights from the clusters
- **Target Marketing:** Use these clusters and segment clusters into different groups.

9. A geospatial research team wants to analyze satellite images to classify different land types (forest, water, urban). What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification(image segmentation)

Step-by-step logic:

- **Data Collection:** Collect the image of satellite images with label data
- **Data Preprocessing:** Extract the pixel values and features from the images.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision and recall.
- **Make Predictions:** Use the best model to predict the class of the satellite images.

10. A streaming service wants to predict which users are likely to cancel their subscriptions. What type of ML problem is this, and what steps would you take to solve it?

Answer:

Problem Type: Classification

Step-by-step logic:

- **Data Collection:** Collect the data such as user frequency of usage, subscription history and content preference
- **Data Preprocessing:** Categorical data handling and null value handling.
- **Data Splitting:** Splitting data into training and testing sets using **train_test_split**.
- **Model Selection:** Select classification models such as logistic regression, k-nearest neighbour, naive bayes etc...
- **Model Training:** Train the data with different models using fit method
- **Model Evaluation:** Evaluate the model using metrics such as accuracy, f1 score, precision and recall.
- **Make Predictions:** Use the best model to predict the user will cancel their subscription or renew it.